# Rahul Yadav *Master's in AI Engineering of Autonomous Systems*

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## Profile

AI Engineering of Autonomous Systems graduate with a multidisciplinary background in computer vision, 3D perception, sensor data, and physical-system modeling. Hands-on experience developing real-time perception pipelines using Python, PyTorch and OpenCV, including LiDAR point-cloud processing, 3D geometry, object detection and quantitative evaluation. Experienced in processing and synchronizing heterogeneous real-world sensor data and analyzing model limitations through field measurements. Combines machine learning experience with a Mechanical Engineering background in vehicle dynamics, simulation, and applied physics, with additional experience in CUDA, and autonomous-system development.

## Education

**Master of Engineering,** *Technische Hochschule Ingolstadt* — 03/2024 – 08/2026 | Ingolstadt, Germany
Artificial Intelligence Engineering of Autonomous Systems, Grade- 1.9 (awaiting)

**Bachelors of Technology,** *K J Somaiya College of Engineering (Mumbai University)* — 08/2017 – 07/2021 | Mumbai, India
Mechanical Engineering, Grade- 1.6

## Master Thesis

**Development of an Data-Driven Vehicle Dynamics Model for XiL Testing Environments** — 12/2025 – Present | Wolfsburg, Germany
*Volkswagen ADMT, MOIA GmbH*
- Developed a LSTM-based vehicle model to estimate vehicle dynamics states from CAN signals.
- Implemented physics-based loss constraints for realistic motion behavior.
- Wrapping this trained network to autoregressively predict vehicle states given the control input, Trained Network -> Vehicle Model.
- Vehicle model achieved 0.90+ R2 score for all signals in open loop evaluation using real world data from proving ground.
- Integrated the vehicle model with Volkswagen's inhouse simulator to run scenarios in closed loop setup with the Self Driving Stack (SDS) in the loop.

## Professional Experience

**Working Student- SimulationX,** *Keysight Inc* — 09/2024 – present | Dresden, Germany
- Developed and enhanced vehicle dynamics models with component-level fidelity.
- Improved energy consumption simulations by modeling tire power loss and suspension behavior.
- Contributed to OpenCRG integration for realistic road surface modeling.

**Junior Engineer,** *Toyo Engineering India* — 08/2021 – 04/2024 | Mumbai, India
- Designed 3D piping layouts and performed isometric analysis using Aveva E3D for IOCL and Nayara Energy projects.

## Projects

**Distributed GPU Training** — 09/2026 – Present
Tech: ML- PyTorch, torchvision, distributed training, model evaluation | MlOps- MLflow, Docker | Infra- NVIDIA CUDA, Vast.ai, Linux | Tools- Git, GitHub, Python
- Developed and containerized a ResNet-18/CIFAR-10 training pipeline with checkpointing, test evaluation, and MLflow experiment tracking.
- Deployed PyTorch DDP on a cloud VM with two RTX 4060 GPUs; achieved 93.8% accuracy and 1.27× speedup over the controlled single-GPU run.
- Published the reproducible Docker image to Docker Hub and persisted models, metrics, and visualizations using mounted storage.

**Visual Perception Pipeline for Autonomous Systems (Using KITTI Dataset)** — 08/2026 – 09/2026
Tech: Python, PyTorch, NumPy, Matplotlib, Jupyter, KITTI, CNNs, U-Net, BEV perception, 3D geometry, sensor calibration.
- Converted raw point clouds into bird's-eye-view representations.
- Designed a lightweight U-Net-style encoder-decoder with multi-scale features and CenterNet-inspired heads to predict vehicle centers, dimensions, and orientations.
- Built robust data loading, augmentation, training, decoding, visualization, and full-dataset evaluation pipelines.
- The model achieved 83.3% precision, 89.0% recall, and an 86.0% F1 score on a held-out validation set, with a 0.14 m mean center-localization error and approximately 27 FPS end-to-end throughput.
- Identified sparse long-range LiDAR observations as the primary limitation, motivating ongoing camera–LiDAR sensor-fusion development.

**Vehicle State Estimation for Autonomous Driving** — 03/2025 – 06/2025
- Developed a vehicle state estimation framework using OBD and ADMA sensor data, sampled at different frequencies..
- Preprocessed and aligned asynchronous sensor data to improve model input consistency and training.
- Implemented and compared RNN, LSTM, and Transformer-based architectures for time-series modeling of vehicle states such as velocity, acceleration, and yaw rate.

- Integrated physics-based constraints into ML models to guide learning and improve generalization across diverse driving scenarios.

**Traffic Sign Recognition (CNN + YOLOv8, GTSRB Dataset)** 08/2025

- Developed a real-time recognition pipeline combining YOLOv8 object detection with a custom PyTorch CNN classifier, achieving 99%+ classification accuracy on GTSRB.
- Built a modular inference pipeline for live video streams and exposed model inference through a FastAPI backend.
- Containerized the application with Docker for reproducible deployment.

## Past Experiences

**Suspension & Vehicle Dynamics Lead,** *Orion Racing India, Formula Student* 11/2018 – 07/2021 | Mumbai, India

- Designed a Z-type anti-roll bar, improving system adjustability by 22%.
- Developed Yaw Moment Diagrams and MATLAB models for traction & launch control.
- Conducted tire data analysis (TTC) to optimize vehicle performance.

## Skills

— **Programming:** Python, C++ • **Deep Learning:** PyTorch, TensorFlow, Scikit-learn • **Computer Vision:** OpenCV, CNNs, YOLOv8, U-Net, Object Detection • **3D Perception:** LiDAR, Point Clouds, BEV Perception, 3D Geometry, Sensor Calibration • **Data & Scientific Computing:** NumPy, Pandas, Matplotlib • **ML Engineering:** Docker, FastAPI, MLflow, NVIDIA CUDA, Linux • **Simulation & Sensors:** CARLA, MATLAB, Simulink • **Version Control:** Git, GitHub

## Publications

**IEEE International Transportation Electrification Conference - India** 2021

- Presented paper at conference which discussed methods to implement Launch Control (LC) in an Formula Student Electric Vehicle.

**International Journal for Research in Applied Science and Engineering Technology** 2022

- Partial Differential Equations (PDE) toolbox from MATLAB was used to perform Finite Element Analysis (FEA) and the results were compared with more common FE software.

## Languages

**English** ●●●●●

**German** ●●●○○
B1-Level

**Hindi** ●●●●●